Happy Number (medium)

Problem Statement

Any number will be called a happy number if, after repeatedly replacing it with a number equal to the **sum of the square of all of its digits, leads us to the number `1`. All other (not-happy) numbers will never reach `1`**. Instead, they will be stuck in a cycle of numbers that does not include `1`.

Given a positive number `n`, return `true` if it is a happy number otherwise return `false`.

Example 1:

Input: 23
Output: true (23 is a happy number)
Explanations: Here are the steps to find out that 23 is a happy number:

$2^2 + 3^2 = 4 + 9 = 13$
$1^2 + 3^2 = 1 + 9 = 10$
$1^2 + 0^2 = 1 + 0 = 1$

Example 2:

Input: 12
Output: false (12 is not a happy number)
Explanations: Here are the steps to find out that 12 is not a happy number:

$1^2 + 2^2 = 1 + 4 = 5$
$5^2 = 25$
$2^2 + 5^2 = 4 + 25 = 29$
$2^2 + 9^2 = 4 + 81 = 85$
$8^2 + 5^2 = 64 + 25 = 89$
$8^2 + 9^2 = 64 + 81 = 145$
$1^2 + 4^2 + 5^2 = 1 + 16 + 25 = 42$
$4^2 + 2^2 = 16 + 4 = 20$

$2^2 + 0^2 = 4 + 0 = 4$

$4^2 = 16$

$1^2 + 6^2 = 1 + 36 = 37$

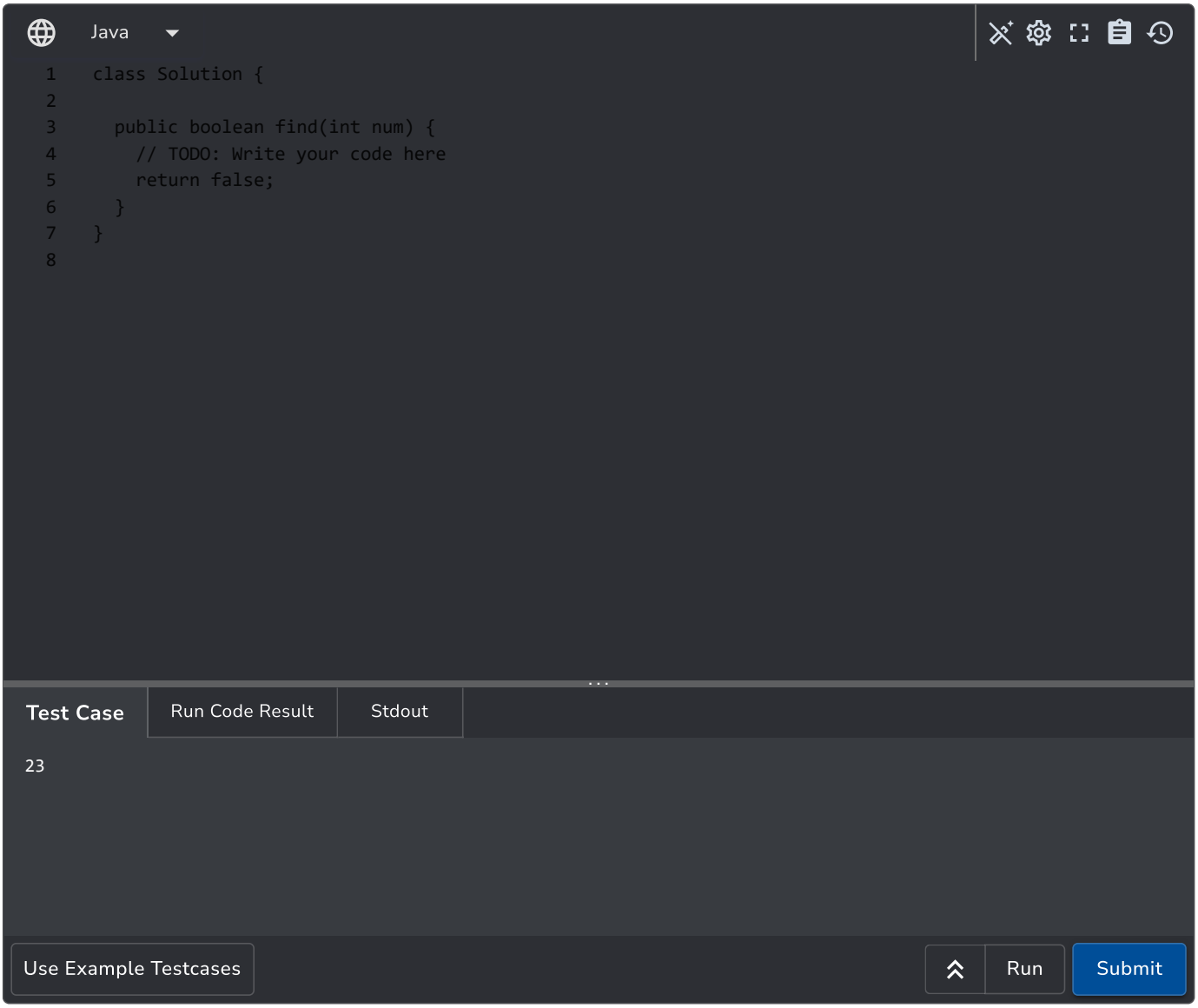
$3^2 + 7^2 = 9 + 49 = 58$

$5^2 + 8^2 = 25 + 64 = 89$

Please note the cycle from 89 -> 89.

Try it yourself

Try solving this question here:



← Previous

Solution: Start Of LinkedList Cycle

Next →

Solution: Happy Number